Experiment No.: 03

Title: Study of Sampling Techniques

Batch: B3 **Roll No.: 16014223083** **Experiment No.: 03**

Aim: To study and understand various sampling techniques used in data analytics and analyze how sampling helps in drawing conclusions about large datasets efficiently.

Resources needed: VSCode, Python(NumPy, Pandas, Scikit-learn), Google Colab, Jupyter Notebook

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Pre Requisite: Data Science, Machine Learning

Theory:

In data analytics, analyzing an entire dataset (population) may be expensive, time-consuming, or impractical. Sampling is the process of selecting a subset of data from a population to represent the whole. A good sample reflects the characteristics of the population and enables accurate analysis.

Types of Sampling Techniques

1. Simple Random Sampling

Each element in the population has an equal probability of being selected.

Advantages

- Easy to implement
  Unbiased selection

Disadvantages

- Not suitable for large or diverse populations

2. Systematic Sampling

Every *k*th element is selected after a random start.

Example
If population size = 1000 and sample size = 100
k = 1000 / 100 = 10
Select every 10th record.

Advantages

- Simple and fast
- Ensures uniform coverage

Disadvantages

- Can introduce bias if data has a pattern

3. Stratified Sampling

Population is divided into strata (groups) based on a characteristic, and samples are taken from each stratum.

Advantages

- More representative
- Reduces sampling bias

Disadvantages

- Requires knowledge of population structure

4. Cluster Sampling

Population is divided into clusters, and entire clusters are selected randomly.

Advantages

- Cost-effective
- Suitable for large geographical datasets

Disadvantages

- Less accurate than stratified sampling

Algorithm / Procedure

1. Load the dataset into the system.
2. Identify population size.
3. Select appropriate sampling technique.
4. Apply sampling method to extract sample.
5. Compare sample characteristics with population.
6. Analyze effectiveness of sampling technique.

Data Set

Iris Dataset/ Sales Dataset/ Savings Dataset
https://www.kaggle.com/datasets/saidaminsaidaxmadov/chocolate-sales

Probabilistic Sampling

- Simple Random Sampling: In simple random sampling, every sales record has an equal chance of being selected. The selection is completely random, so there is no bias.
- Stratified Sampling: In systematic sampling, we select every k-th record from the dataset after deciding on a fixed interval.
- Systematic Sampling: In stratified sampling, the dataset is divided into groups called strata based on a specific feature, and samples are taken from each group.
- Cluster Sampling: In cluster sampling, the dataset is divided into clusters, and one or more entire clusters are selected randomly.

Non-Probabilistic Sampling

- Convenience Sampling: In convenience sampling, samples are selected based on ease of access rather than randomness
- Judgmental (Purposive) Sampling: In judgmental sampling, samples are selected based on specific conditions or expert judgment.
- Quota Sampling: In quota sampling, a fixed number of samples is selected from each category.

```
import pandas as pd
df = pd.read_csv(r"/content/chocolate-sales/Chocolate Sales (2).csv")
df
```

	Sales Person	Country	Product	Date	Amount	Boxes Shipped
0	Jehu Rudeforth	UK	Mint Chip Choco	04/01/2022	$5,320.00	180
1	Van Tuxwell	India	85% Dark Bars	01/08/2022	$7,896.00	94
2	Gigi Bohling	India	Peanut Butter Cubes	07/07/2022	$4,501.00	91
3	Jan Morforth	Australia	Peanut Butter Cubes	27/04/2022	$12,726.00	342
4	Jehu Rudeforth	UK	Peanut Butter Cubes	24/02/2022	$13,685.00	184
...	...	...	...	...	...	...
3277	Karlen McCaffrey	Australia	Spicy Special Slims	17/05/2024	$5,303.58	354
3278	Jehu Rudeforth	USA	White Choc	07/06/2024	$7,339.32	121
3279	Ches Bonnell	Canada	Organic Choco Syrup	26/07/2024	$616.09	238
3280	Dotty Strutley	India	Eclairs	28/07/2024	$2,504.62	397
3281	Karlen McCaffrey	India	70% Dark Bites	23/05/2024	$5,915.87	355

3282 rows × 6 columns

```
# PROBABILISTIC SAMPLING TECHNIQUES

# 1.Simple Random Sampling
simple_random_sample = df.sample(n=100, random_state=42)
print("Simple Random Sample size:", len(simple_random_sample))
print("\n")

# 2.Systematic Sampling
k = 10
systematic_sample = df.iloc[::k]
print("Systematic Sample size:", len(systematic_sample))
print("\n")

# 3.Stratified Sampling
stratified_sample = (
    df.groupby("Country", group_keys=False)
      .apply(lambda x: x.sample(frac=0.2, random_state=42))
)
print("Stratified Sample size:", len(stratified_sample))
print(stratified_sample["Country"].value_counts())
print("\n")

# 4.Cluster Sampling
import random
clusters = df["Country"].unique()
```

```
selected_cluster = random.choice(clusters)
cluster_sample = df[df["Country"] == selected_cluster]
print("Selected Cluster:", selected_cluster)
print("Cluster Sample size:", len(cluster_sample))
# NON-PROBABILISTIC SAMPLING TECHNIQUES

# Clean 'Amount' column by removing '$' and ',' and converting to float
df["Amount"] = df["Amount"].replace({r'\$': '', r',': ''}, regex=True).astype(float)

# 1.Convenience Sampling
convenience_sample = df.head(100)
print("Convenience Sample size:", len(convenience_sample))
print("\n")

# 2.Judgmental (Purposive) Sampling
judgmental_sample = df[df["Amount"] > df["Amount"].mean()]
print("Judgmental Sample size:", len(judgmental_sample))
print("\n")

# 3.Quota Sampling
quota_sample = (
    df.groupby("Country", group_keys=False)
      .apply(lambda x: x.sample(n=20, random_state=42))
)
print("Quota Sample size:", len(quota_sample))
print(quota_sample["Country"].value_counts())
```

Output:

```
Simple Random Sample size: 100


Systematic Sample size: 329


Stratified Sample size: 656
Country
Australia      123
India          110
USA            107
UK             107
Canada         105
New Zealand    104
Name: count, dtype: int64


Selected Cluster: New Zealand
Cluster Sample size: 519
```

```
Convenience Sample size: 100


Judgmental Sample size: 1427


Quota Sample size: 120
Country
Australia      20
Canada         20
India          20
New Zealand    20
UK             20
USA            20
```

Outcomes:

CO2: Implement the statistical methods to analyse and interpret the data

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Conclusion:

The Chocolate Sales dataset was used to implement various probabilistic and non-probabilistic sampling techniques, demonstrating how representative samples can be extracted efficiently from large sales data.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:
Websites:

1. https://medium.com/@ayanchowdhury00/sampling-techniques-and-its-implementation-in-python-487995b9399c